

Kris Huynh

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EXPERIENCE

New Jersey City University – NASA Funded AI Research Project

Jersey City, NJ

Undergraduate Research Assistant – Lead Student Project

Sep 2025 - Present

- Built a **computer vision benchmarking pipeline** comparing classical and deep learning edge detection methods (**Sobel, Canny, HED, DexiNed**) using **Python, PyTorch, and OpenCV**.
- Evaluated performance on **spacecraft imagery datasets** using **precision, recall, Boundary F1 score, and Chamfer distance metrics**.
- Demonstrated that **deep learning models outperform classical methods**, with **DexiNed achieving F1 = 0.705** on BIPED and **HED achieving F1 = 0.681** on spacecraft imagery.
- Developed **CAD wireframe alignment analysis** to measure geometric accuracy of predicted edges for spacecraft pose estimation applications.

New Jersey City University

Jersey City, NJ

Undergraduate Research Assistant

June 2025 - Aug 2025

- Improved fraud detection recall to **87.8%** on highly imbalanced data (0.17% fraud) by applying **SMOTE and class-weight techniques** across Logistic Regression, Random Forest, XGBoost, and LightGBM.
- Achieved **0.97 ROC-AUC** by evaluating model performance (Precision, Recall, F1-score), identifying **XGBoost as the optimal trade-off between fraud detection and false positives**.
- Developed a **machine learning pipeline (Python, Pandas, Scikit-learn)** for data preprocessing, model training, feature importance analysis, and hyperparameter tuning.

PROJECTS/ FREELANCE

Retail Analytics (Amazin Choices LLC) | Project-based

Jan 2026 - Present

- Cleaned and validated multi-source retail data (5+ platforms) using **Python, BigQuery, and dbt**, resolving **70+ data quality issues** and improving data reliability for downstream analytics.
- Performed **advanced SQL analysis** to uncover insights on revenue trends, customer behavior, and marketing performance.
- Designed and developed interactive Tableau dashboards** (Revenue Trends, Product Profitability, Marketing ROI), enabling real-time business monitoring and supporting data-driven decision-making.

AI-Powered Marketing Automation System | [Github](#)

Feb 2026

- Designed and implemented an **end-to-end AI-powered** marketing automation pipeline using **n8n and Gemini LLM**
- Automated personalized **email campaign generation and delivery**, reducing manual effort and improving campaign efficiency across customer segments.

Predicting On-Time Shipping with Machine Learning | [Github](#)

Sep 2025

- Built a **machine learning pipeline** to predict delivery delays using **Logistic Regression and SVM** on a dataset of **43K+ shipping records**.
- Engineered predictive features including **route distance (Haversine), traffic patterns, weather conditions, and delivery agent performance**, achieving **80% prediction accuracy**.

Data Warehouse and Integrity | [Github](#)

Mar 2025

- Designed and implemented **Bronze–Silver–Gold warehouse architecture** in PostgreSQL for ERP/CRM data.
- Built ETL pipelines** to extract, clean, and transform data; implemented **40+ data quality checks** to ensure accuracy and consistency.

TECHNICAL SKILLS

Programming & Databases:	Python(pandas, numpy, PyTorch), Javascript, SQL (PostgreSQL, BigQuery)
Visualization:	Tableau, Excel (PivotTables, VLOOKUP)
Data & Analytics:	Data modeling, ETL pipelines, Data validation, Data warehousing
Tools:	Git, GitHub, Jupyter, dbt
AI/Automation:	n8n, LLM API integration, prompt engineering

EDUCATION

New Jersey City University, Jersey City, New Jersey

Exp: May 2026

Bachelor of Science in Computer Science - GPA: 3.9